Moein Bagheri

Building AI that listens · Speech & Audio AI

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Computer Engineering student on the AI track, working on machine learning for speech, audio and language. Open to AI roles in NLP, speech and signal processing.

EDUCATION

B.Sc. in Computer Engineering, University of Isfahan — Sep 2022 – Expected 2027

- GPA **17.26/20** overall · **18.5/20** over the last three semesters and in specialized courses · AI major, IT minor
- Selected courses: Signals & Systems (20) · Language & Speech Processing (20) · Design of Algorithms (20) · Data Mining (19.75) · Probability & Statistics (19.25) · Applied Linear Algebra (18.87) · Deep Learning (18.5)
- Final-year project (in progress): Persian–English code-switched speech recognition · Supervisor: Dr. S. F. Noorbehbahani

NODET (SAMPAD) High School, Yazd — Diploma

Iran's schools for exceptional talents, entered through a national exam

SELECTED PROJECTS

Speech Emotion Recognition — PyTorch · librosa · 2026

- Trained plain and residual 1D CNNs on MFCC features to recognize four emotions in IEMOCAP (4,901 clips, 10 speakers).
- Speaker-independent evaluation showed what a random split hides: accuracy fell from 65.1% to 58.0% on unseen speakers, and a higher score could come from a model that never predicted "happy".

Local Research Assistant (RAG) — Ollama · FAISS · Streamlit · 2026

- Built a local assistant over four retrieval papers (REALM, DPR, RAG, FiD) that cites the page behind each answer.
- Retrieval raised answers containing the reference fact from 8% to 75% on a 12-question test; a six-tool router with a keyword fallback handled all 12 requests.

Persian Search Engine *(with S. Heydari)* — Python · Hazm · 2026

- Implemented an inverted index, Boolean search, TF-IDF and BM25 by hand over 100 Persian thesis records.

Deep ECG Classification — TensorFlow · 2025

- Trained GRU networks for five-class heartbeat classification on MIT-BIH (93.3% accuracy); per-class analysis showed the headline hid 6% recall on supraventricular beats.

More on GitHub: n-gram melody completion · neural style transfer · YOLOv8 gesture detection · DiaTech product discovery

EXPERIENCE

E-Commerce Lab, University of Isfahan — Summer 2025

Business Development Intern · Supervisor: Dr. S. F. Noorbehbahani

- Conducted market research for two early-stage startup concepts: an online physiotherapy platform and a personalized children's storybook service.
- Took part in the hands-on work of setting up the Computer Innovation Center's workspace ahead of its opening.

LEADERSHIP & HONORS

- Founded and ran Bartarinha, a team math competition between Yazd's NODET boys' and girls' schools (2017–18).
- Hosted the Computer Engineering department's welcome ceremony for new students, at the faculty's invitation.
- Third place in piano, Yazd students' cultural and arts competitions (2019).

SKILLS

Programming	Python, Java, C/C++, SQL, JavaScript, Git
Speech & ML	PyTorch, TensorFlow/Keras, scikit-learn, librosa, OpenCV, Ultralytics YOLO
NLP & retrieval	Hazm, NLTK, FAISS, Ollama, retrieval-augmented generation, Streamlit
Languages	Persian (native), English (fluent)